Elastic Coefficients Directly from RealQM

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Abstract

A macroscopic elastic constant is conventionally obtained from a microscopic theory by homogenisation: pose a two-scale expansion, solve a cell problem, average a stress field, and take a limit in which the microscale vanishes. This paper makes the point that for *equilibrium* coefficients none of that is necessary. An elastic modulus is a second derivative of the ground-state energy with respect to a macroscopic deformation; the deformation can be imposed exactly on a periodic cell as a boundary condition; and the energy is then a single number. No field is averaged and no limit is taken — a scalar is computed and differentiated. We carry this out in RealQM, a continuum formulation of quantum mechanics in which an N-electron system is N non-overlapping unit charge densities on ordinary three-dimensional space with free interfaces, so that the microscopic theory is itself a continuum theory and there are no particles to coarse-grain. Equilibrium densities and bulk moduli are computed for the five alkali metals; the error falls monotonically down the group and reaches zero at caesium, while the standard treatment's does not. Lithium fails outright on the literature's empty-core radius, and we locate the failure in that input rather than in the construction: the empty-core radii of the other four alkalis are a single constant times the pseudo-kernel radius this theory fixes from the measured ionization energy, to under one per cent, and applying that constant to lithium — which is excluded from it, so this is a prediction — takes its modulus from 4.3 times too stiff to 7% high. One constant for the series replaces one radius per element. The route is then checked against the mechanics it bypasses: the pressure and the modulus obtained from the stress balance $\nabla \cdot T = -\rho \nabla V$ agree with those obtained by differentiating the energy, to 2×10^{-4} and 5×10^{-4} after extrapolation. The *form* of the macroscopic law is verified rather than assumed — the energy is observed to be quadratic in the strain out to a few per cent — and that same test locates the method's accuracy limit at the *order* of the derivative taken: energies to a few per cent, equilibrium densities to one, moduli to ten or twenty, and the first anharmonic coefficient not at all. Finally the surface energy is computed, as the one elastic-body quantity requiring no scale-separation assumption, since in this formulation a surface is a free boundary rather than a truncated lattice. What the route does not reach is any dissipative coefficient, and the reason is sharp: it reaches a coefficient if and only if that coefficient is a derivative of an energy.

1 The problem, and why averaging is not needed for it

Deriving the equations of continuous media from a microscopic description of matter is the content of the part of Hilbert's sixth problem that Hilbert attributed to Boltzmann, and the classical route to it — particles, then a kinetic equation, then Euler and Navier–Stokes — was

carried through for hard spheres by Deng, Hani and Ma in 2025 [1]. That route reaches a *dilute gas* by construction, since the Boltzmann–Grad scaling sends the volume fraction to zero [2], so solids lie outside it. The wider programme of which this paper is a part takes the complementary route and is set out elsewhere [9]; the present paper isolates the part of it that is finished, namely the equilibrium coefficients, and reports what they do and do not deliver.

The starting point is that the usual machinery is aimed at a harder problem than the one at hand. Homogenisation exists to extract a macroscopic law from a microstructure whose fields oscillate, and it earns its two-scale expansion, its corrector problem and its limit. An *equilibrium* coefficient asks for less. It is a derivative of a scalar — the ground-state energy — with respect to a parameter which is already macroscopic, and a derivative of a scalar needs no averaging of anything. The whole content of this paper is that taking the derivative directly is both available and checkable, and that its limits are then visible rather than hidden in an asymptotic estimate.

The microscopic theory used here is RealQM [4, 5], in which an N-electron system is represented by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy is Coulomb's,

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2 \Big) dx + \tfrac{1}{2}\sum_{i\neq j} \iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \tag{1}$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$, and the ground state minimises it. The passage made below is from a continuum on the atomic scale a_0 to a continuum on the laboratory scale L, with small parameter

$$\varepsilon = a_0/L, \tag{2}$$

which is carried throughout as an *assumption* — that the macroscopic state is uniform across a cell — rather than as an asymptotic analysis. That is the one place where a scale separation enters, and it is stated here so that it is not mistaken for a derivation. Two features matter for what follows. It is already a continuum theory on ordinary three-dimensional space, so there are no particles to coarse-grain and no statistical ensemble anywhere in the construction. And its stationarity conditions are a stress balance, which is what makes the check of Section 3 possible at all.

2 The static stress balance

The check performed later needs the mechanics the direct route bypasses, so it is written down first. Two identities do the work, and both are the chain rule.

The localisation energy is the von Weizsäcker functional. Writing $\rho_i = \psi_i^2$ with $\psi_i \geq 0$, the chain rule gives $\nabla\rho_i = 2\psi_i\nabla\psi_i$, hence

$$\tfrac{1}{2}\int |\nabla\psi_i|^2\,dx \;=\; \frac{1}{8}\int \frac{|\nabla\rho_i|^2}{\rho_i}\,dx \;=\; T_{\mathrm{vW}}[\rho_i], \tag{3}$$

in units $\hbar = m = 1$. An identity, not a model.

Its variation is the quantum potential. With $u = \sqrt{\rho}$,

$$\frac{\delta T_{\mathrm{vW}}}{\delta \rho} \;=\; -\tfrac{1}{2}\frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} \;=\; Q, \tag{4}$$

Bohm's quantum potential — obtained here not by analogy but as the variation of a term (1) writes down for an unrelated reason, namely the cost of confining a charge to a region.

The theory as used is a hydrostatics. Stationarity of (1) is the statement that $V + Q$ is constant on each domain, which is the $v = 0$ case of the Madelung pair

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho v) = 0, \tag{5}$$

$$\frac{\partial v}{\partial t} + (v\cdot\nabla)v = -\nabla\big(V + Q\big). \tag{6}$$

Equation (5) is conservation of charge and (6) is a momentum balance whose stress is supplied by (4). *RealQM is therefore already the equilibrium of a continuum mechanics, and the constitutive law is not postulated — it is the variation of the energy the theory was written with.* This is the sense in which the derivation asked for by Hilbert is, on this route, partly a matter of recognition rather than of proof.

And the interfaces carry a surface tension. The free boundary between two domains contributes an energy per unit area, so the multiphase structure is a continuum mechanics *with surfaces*, which is what dense matter requires and what a kinetic theory has no room for [3].

The stress, written down. With $u = \sqrt{\rho}$ and $Q = -\frac{1}{2}\nabla^2 u/u$, the chain rule alone gives

$$\rho\,\nabla Q \;=\; \nabla\cdot T, \qquad T_{ij} \;=\; -\tfrac{1}{2}\big[\,u\,\partial_i\partial_j u - \partial_i u\,\partial_j u\,\big] \;=\; -\tfrac{1}{4}\partial_i\partial_j\rho + \frac{\partial_i\rho\,\partial_j\rho}{4\rho}, \tag{7}$$

with no δ_{ij} term; in one dimension this is $\sigma = \frac{1}{4}[(\rho')^2/\rho - \rho'']$. The static Euler–Lagrange equation of the model is $Q + V = \lambda$ with $V = -K + P_{\mathrm{other}}$, so the $v = 0$ momentum balance is the statement

$$\nabla\cdot T \;=\; -\rho\,\nabla V, \tag{8}$$

and integrating its z-component over a half space $\{z > z_{\mathrm{cut}}\}$ turns it into a statement about forces: the flux of quantum stress through the cut equals the net electrostatic force on the charge beyond it. The left-hand side is a surface integral of a quantity carrying a *third* derivative of the density; the right-hand side is a volume integral of Coulomb forces. Nothing makes them equal except the identity.

Equation (8) is the statement to be checked against: a stress balance which the model satisfies pointwise, with no averaging and no approximation beyond the chain rule.

3 Direct upscaling: no averaging required

Consider what an elastic modulus actually is. It is a second derivative of the ground-state energy with respect to a macroscopic deformation,

$$C_{ijkl} = \frac{1}{V}\frac{\partial^2 E}{\partial\epsilon_{ij}\,\partial\epsilon_{kl}}, \qquad B = V\frac{\partial^2 E}{\partial V^2}, \tag{9}$$

and the deformation ϵ is a *macroscopic* quantity that can be imposed on a periodic cell exactly, as a boundary condition: one changes the shape or the volume of the cell. RealQM then computes the ground state in that cell and returns a single number, its energy. The modulus is the curvature of that number as a function of the imposed deformation.

Nothing has been averaged over the microscale. No stress field is integrated, no two-scale expansion is posed, no corrector problem is solved. A scalar is computed exactly and differentiated with respect to a macroscopic parameter. The only use made of the cell is that the macroscopic state is uniform across it — which is the content of $\varepsilon \ll 1$ in (2), carried as an assumption rather than as an asymptotic analysis.

What this reaches. Every coefficient of continuum mechanics that is a derivative of the ground-state energy with respect to a macroscopic control parameter:

- the full elastic tensor and the bulk, shear and Young moduli;
- the equation of state $P(V) = -\partial E/\partial V$, and with it the speed of sound;
- the force constants $\partial^2 E/\partial u_a \partial u_b$ and hence the phonon spectrum, and through it the thermal expansion in the quasi-harmonic treatment;
- the dielectric and piezoelectric response, as derivatives with respect to an applied field;
- surface and interface energies, as differences of computed energies.

That is the whole of equilibrium continuum mechanics. It requires no new machinery — only the ground-state solver the programme already has.

A worked example that already exists, across five materials. The alkali metals have been computed this way on a Wigner–Seitz cell, one domain per ion filling the cell with the normal derivative vanishing at the face by symmetry — which is RealQM's own prescription and coincides with a construction in use since 1933. Here r_s is the *Wigner–Seitz radius*, the radius of the sphere whose volume is the volume per valence electron,

$$\tfrac{4}{3}\pi r_s^3 = \frac{V}{N}, \qquad \text{so} \qquad n = \left(\tfrac{4}{3}\pi r_s^3\right)^{-1},$$

quoted in Bohr radii: with one valence electron per ion, each ion sits at the centre of a neutral sphere of radius r_s holding exactly one electron, and $\psi'(r_s) = 0$ on its face — which is where RealQM's free interface and the 1933 condition coincide. Compressing the metal is decreasing r_s, so $E(r_s)$ *is* the equation of state: it is minimised for the equilibrium density and differentiated twice for the modulus, with $V = \frac{4}{3}\pi r_s^3$ converting the r_s derivatives into the volume derivatives of (9). It should not be confused with the lattice constant, which for sodium is about twice as large, nor with the empty-core radius r_c below, which is the pseudopotential parameter and roughly half as large. In the table immediately below, each element brings one parameter, its Ashcroft empty-core radius, taken from the literature and not fitted here; the paragraph after next removes that per-element input in favour of a single constant for the whole series. Across the series r_s varies by 70% and B by a factor of six, so what is being tested is a trend and not a single match.

	equilibrium r_s (a_0)			bulk modulus (GPa)		
	RealQM	standard	measured	RealQM	standard	measured
Li	2.40	2.86	3.25	49.6	23.9	11.6
Na	3.77	3.87	3.93	9.1	8.0	6.3
K	4.82	4.67	4.86	3.6	4.0	3.1
Rb	5.20	4.95	5.20	2.7	3.2	2.5
Cs	5.62	5.28	5.62	2.0	2.5	2.0

The RealQM column keeps the same electrostatics as the standard one and differs only in the kinetic energy — the cell state's own $\langle T \rangle$ in place of a band-filling term, and no exchange. Macroscopic material constants, from Coulomb geometry.

The shape of the error is the result, not its size. RealQM's error falls *monotonically down the group* and reaches zero at caesium, on both quantities: the deviation in r_s runs -26%, -4.1%, -0.8%, 0%, 0% and in B runs $+327\%$, $+44\%$, $+18\%$, $+6\%$, 0%. The standard column does nothing of the kind — its r_s error *worsens* down the group, from -1.5% at sodium to -6.0% at caesium, and its bulk modulus sits at a nearly constant $+25$ to $+29\%$. A model that merely happened to fit would not produce a sequence ordered by atomic number. Setting lithium aside, RealQM is the more accurate of the two columns on both quantities: 1.2% against 4.1% on r_s, and 17% against 27% on B.

Lithium's failure is the empty-core input, and one constant repairs it. Lithium's modulus above comes out 4.3 times too stiff, and lithium is also the element that stands outside the pattern in this model's alkali-dimer work, where the core radius fitted to the measured ionization energy exceeds the ionic radius by $0.56\,a_0$ against 0.1–0.2 for every other alkali. The same element falling outside two independent calculations is a fact about lithium, not scatter — and the two facts turn out to be one fact, which identifies the culprit.

That dimer work fixes, for each alkali, a pseudo-kernel radius R by requiring that one electron outside R on a charge $+1$ kernel reproduce the element's *measured ionization energy*. R is therefore determined by an atomic measurement, with no solid-state datum anywhere in it. Comparing it with the empty-core radii used above,

$$\frac{r_c}{R} \;=\; 0.773,\ 0.767,\ 0.777,\ 0.759 \qquad \text{for Na, K, Rb, Cs,}$$

a single constant $C = 0.7688$ with a standard deviation of 0.008 — under one per cent across four elements spanning a factor 1.5 in R. It is also the better of the two available proportionalities: against the measured ionic radius the same ratio scatters at 2.8%, four times worse, so the radius this theory determines predicts the core better than the measured ionic radius does.

Lithium is excluded from that mean, where its ratio is 0.530. So setting $r_c = CR$ for lithium is a *prediction*, drawing on lithium's own ionization energy and four other elements' core radii, and on no property of lithium's solid. It gives $r_c = 1.537\,a_0$ in place of the literature's 1.06, and with it

	R(IE)	$r_c = CR$	equilibrium r_s (a_0)		bulk modulus (GPa)	
	(a_0)	(a_0)	RealQM	measured	RealQM	measured
Li	1.999	1.537	3.47	3.25	12.5	11.6
Na	2.161	1.661	3.75	3.93	9.2	6.3
K	2.792	2.147	4.83	4.86	3.6	3.1
Rb	2.972	2.285	5.14	5.20	2.8	2.5
Cs	3.295	2.533	5.69	5.62	1.9	2.0

Lithium's modulus goes from 4.3 times too stiff to 7% high, and its equilibrium radius from 26% low to 7% high. Over the whole series the mean absolute error falls from 6.2% to 2.9% in r_s and from 79% to 17% in B; over the four elements that set C it is a wash, 1.2% $\to$ 1.9% and 17% $\to$ 19%, so the entire gain is lithium. The repair is not a knife-edge either: carrying C from 0.70 to 0.84 moves lithium's r_s through 3.16 $\to$ 3.79 and its B through 17.7 $\to$ 8.9, crossing the measured values smoothly, and the standard deviation of C propagates to some 3% on B.

So the one-electron-per-cell picture was not what failed at lithium. The empty-core radius was, and the theory's own radius — fixed by an ionization energy — is the better input. What the series needs is not five parameters but one.

The constant is monovalent. Carried to divalent magnesium it does not transfer: the same construction, one electron on a +1 kernel matched to the first ionization energy, gives $r_c = 0.813$ against the literature's 1.39, low by 42%. A valence-matched pseudo-atom — a +2 kernel with two electrons and the self-repulsion reduced by $\frac{1}{2}$, matched to the sum of the first two ionization energies — gives 1.291, low by 7%, which is far better but is a different construction on which C was never fitted. One constant for all therefore holds within a valence column, and whether each column has its own constant is open.

Three things should be kept attached to the table. The calculation is not parameter-free, but the paragraph above reduces its freedom to one number for the whole series — a single constant C, plus each element's measured ionization energy, in place of an empty-core radius per element. A bulk modulus is a second derivative, so it amplifies the model's error in the way H_2's binding energy amplifies the error on its total — which is why the percentages are large where they are large. And the zeros at rubidium and caesium mean agreement to the two or three significant figures the measured values carry, not exactness. The point of the example is not that the numbers are good. It is that they exist, were obtained without averaging, required no ingredient the theory did not already have, and vary with the element in a way the theory did not get to choose.

The route is checked against the stress balance it replaces. The objection to (9) is not that the energy is hard to compute but that differentiating it *bypasses* the mechanics: no stress is ever formed, so nothing guarantees that the number differentiated is the number a force balance would have produced. That is now checked, and it had to be, because the two are the same physics.

Take the one-dimensional analogue of the Wigner–Seitz construction above: a chain of period a, one kernel of charge +1 and one electron per cell, each electron on its own cell with the free interfaces at the cell faces by symmetry — RealQM's non-overlap, and the same prescription the sodium calculation uses. In equilibrium the total momentum flux through any plane is the same number, and that number is the pressure. So the pressure and the modulus can each be

computed twice, once by differentiating the energy and once from the stress,

$$\sigma \;=\; \underbrace{T_{zz}(\text{face})}_{\text{localisation stress}} \;+\; \underbrace{F_{\text{Coul}}(\text{face})}_{\substack{\text{electrostatic force of everything left}\\ \text{of the face on everything right}}} \qquad \text{against} \quad -\frac{dE}{da}, \qquad a\frac{d\sigma}{da} \quad \text{against} \quad a\frac{d^2E}{da^2},$$

with no ingredient shared between the two sides beyond the converged ground state. They agree, and the agreement improves under refinement:

relative difference	$n = 41$	$n = 61$	$n = 81$	$n = 101$	extrapolated
pressure, σ against $-dE/da$	2.3%	1.6%	1.2%	0.96%	2.2×10^{-4}
modulus, $a\,d\sigma/da$ against $a\,d^2E/da^2$	3.3%	2.2%	1.6%	1.3%	4.5×10^{-4}

Two features of the convergence are worth recording. It is *first order* in the mesh, and the reason is the one Section 2 identified: the stress is evaluated *at* the free interface, where this construction is first-order accurate although it is fourth-order in the bulk. The free boundary is again the accuracy-limiting feature. And neither lattice sum is responsible — the Coulomb flux is converged by the sixth neighbour and the lattice energy by the tenth, because both are summed as *neutral* cells, each of which is inversion symmetric about its kernel and therefore carries no dipole, leaving a quadrupolar $1/d^3$ tail. Summed term by term the one-dimensional Coulomb sums diverge logarithmically; summed as neutral units they converge at once.

What this establishes is modest and worth having: the scalar that (9) differentiates is the scalar a stress balance returns. The direct route is a shortcut through the mechanics, not a detour around it.

The form, verified, and where the route's accuracy actually runs out. The objection to (9) that cannot be answered by computing more coefficients is that it presupposes the *form*: E quadratic in the strain, which is Hooke's law put in rather than derived. The programme's answer is to establish the form computationally, and that has now been done, with a result sharper than expected in one direction and worse in another.

On the periodic chain, straining the cell about a reference state and fitting a quadratic on $|\epsilon| \le 0.05$ alone, the energy then *extrapolates* as a quadratic out to about $|\epsilon| \approx 0.06$ within 5% of the quadratic term, with the cubic contribution rising smoothly to 10% at $|\epsilon| = 0.10$ and 20% at 0.22. So the quadratic form is not assumed but observed, over a range of strain, with the departure appearing where anharmonicity should.

The quantitative version of the same test is less kind, and it bounds the direct route in a way worth stating. The cubic coefficient is not a free shape parameter: it fixes $B' = dB/dP$, the pressure derivative of the bulk modulus, which is measured and which for the alkalis is close to 4. Computed from the same energy curves that gave the table above, by finite differences and with no equation of state fitted, the model returns

$$B' \;=\; -0.71,\ 0.10,\ 1.35,\ 1.92,\ 2.63 \qquad \text{for Li, Na, K, Rb, Cs,}$$

against a measured value near 4 throughout. The familiar monotone improvement down the group is there again, and caesium is again the best, but even caesium is a third low and lithium comes out with the wrong sign — a bulk modulus that *falls* under pressure, which no alkali does.

The lesson is not that the route fails but that its accuracy is bounded by the *order of the derivative* rather than by the kind of coefficient. The energy of these cells is good to a few per

cent, the first derivative gives equilibrium densities to about one per cent for the heavy alkalis, the second gives bulk moduli to ten or twenty per cent, and the third is gone. The claim above, that this route reaches every coefficient which is a derivative of an energy, should be read with that ladder attached, because each differentiation spends accuracy the model does not have to spare.

Reaching a body, and the one quantity that needs no scale separation. Coefficients are not yet the statics of an elastic body. For that one needs a macroscopic stress field $\sigma(x)$ over the body, hence that the material at each point behaves like a uniformly strained cell — the *Cauchy–Born* hypothesis. Granted it, elastostatics follows in the ordinary way: minimise $\int_\Omega W(\epsilon(u))\,dx$ less the work of the loads, with W the cell energy computed above, and the Euler–Lagrange equation of that functional is $\nabla \cdot \sigma = f$ with the given boundary conditions. Nothing further is required of RealQM. But Cauchy–Born is assumed here, exactly as $\varepsilon \ll 1$ is assumed in (2), and it is known to fail at surfaces, defects and crack tips — which is to say at most of the problems elasticity is wanted for.

That failure is where this formulation has an advantage rather than a handicap. In a lattice model a surface can only be made by truncating the lattice. Here a surface *is* a free boundary, the object the theory is built on, and a surface energy is therefore a difference of computed energies needing no scale separation at all. On the finite chain, with the interior interfaces at the cell midpoints and the two end domains left to run out into vacuum and find their own extent, $E(N) = N\,e_{\mathrm{bulk}} + 2\gamma$ gives

$$\gamma = 0.0416 \text{ Ha} = 1.13 \text{ eV per end},$$

and the test that makes this a surface property rather than a fitted intercept is that γ is *independent of* N — flat to 3×10^{-7} from $N = 5$ onwards. It converges at second order in the mesh, and the bulk energy read off the finite chains agrees with the periodic calculation, computed by different code, also at second order. So the free boundary carries a surface energy, it is stable against the size of the system, and it was obtained without the hypothesis that the rest of the elastic programme rests on.

4 Where the direct route stops

And where it stops, which is the informative part. The direct route reaches a coefficient if and only if that coefficient is a derivative of an energy. *Viscosity is not.* A viscous stress is dissipative: the work done against it is not stored, so there is no E(strain rate) to differentiate, and no variational principle whose minimiser contains η. This is exactly why the standard treatment reaches for Green–Kubo, which extracts a transport coefficient from equilibrium fluctuations and therefore needs an ensemble.

So the division of the problem is sharp, and it is not an artefact of the method chosen:

	conservative coefficients	dissipative coefficients
examples	elastic tensor, $P(V)$, phonons	viscosity, thermal conductivity
nature	derivative of ground-state energy	transport, no energy functional
this route	**direct, no averaging**	**not reached**

Equilibrium continuum mechanics follows from RealQM by differentiating computed energies. Dissipation does not follow at all, and is the whole of what remains open (Section 4).

What is given up, stated plainly. The direct route delivers the *coefficients* of the macroscopic laws, not their *form*. Writing (9) presupposes that E is quadratic in the strain near equilibrium — that is, it assumes Hooke's law rather than deriving it. Homogenisation would instead produce linear elasticity as the macroscopic limit, form and coefficient together, and that is a genuine advantage of the formal route which should not be waved away.

Two things reduce the force of the objection. The form can be established computationally: compute $E(\epsilon)$ across a range and verify that it is quadratic, and that the departure from quadratic sets in where nonlinear elasticity is observed to set in. That is a demonstration and not a theorem. And a demonstration is the kind of evidence this programme is built to give — its standing claim is that its assertions are testable *because* computable, and reaching instead for an asymptotic theorem would borrow precisely the sort of authority it declines to grant to the standard formalism elsewhere. The formal route remains worth having; it is not what the next computation should wait for.

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